Shashikant Yadav

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Academic Qualifications

Year	Degree	Institute	CPI/%
2021 - 2025	B.Tech- Mechanical Engineering	Indian Institute of Technology Kanpur	7.1/10
2021	Grade XII (CBSE)	Prince Uch Madhyamik Vidyalaya, Sikar	92.6%
2019	Grade X (CBSE)	Oasis Sainik School, Kalibanga	89.4%

Patents

- High Actuation Frequency Shape Memory Alloy Based Rotary Actuator *(Indian Patent : Draft)* 2024
- Efficient Container Weighing and Packaging System for Retail *(Indian Patent : Draft)* 2024

Research Work

Undergraduate Researcher | SMSS Lab, IIT Kanpur | *Mentor: Prof. Bishakh Bhattacharya, Department: ME, IIT Kanpur* — *May'23 - Present*

Objective	• Design and **development of novel rotary actuation system** powered by shape memory alloy coil driven mechanism • Size optimization of **shape memory alloy based rotary actuator** by **50%** without compromising outputs in existing system
Approach	• Performed experimental study to determine **correlation of actuation force** of SMA coil with respect to its length/pitch • Developed a **new design framework** by critically analyzing and identifying the areas of improvements in existing system
Result	• **Reduced size by 79.6%** by fabricating compact SMA-powered cam follower mechanism using Addictive manufacturing • Proposed product serves as **better alternative to conventional actuation system** and aims to improve **cost-benefit ratio** • New actuator design, resulted in **Publication as Research Paper and Patent** for its innovative and highly efficient

Professional Experience

Full Stack Game Developer | Vizuara, Cambridge — *Feb'23 - Present*

Objective	• Developed **web-based interactive mini-games for students** to visualize activities mentioned in their school textbooks
Approach	• Utilized the Unity game engine and C# language, emphasizing on **performance optimization** and **code reusability** • Created high-end game models for enhanced visuals and optimized **user experience** to maximize player engagement
Result	• Designed **15+** web-based games for children, incorporated graphics, and delivered an exceptional playing experience

Key Projects

Slope Delivery Bot | Course Project | *Mentor: Prof. Mohit Law, Department: ME, IIT Kanpur* — *Jan'23 - May'23*

Objective	• Engineered a **sustainable bot** utilizing recycled materials to **deliver packages** on inclined and rough surfaces
Mechanism	• Performed detailed force and torque calculations to determine the **optimal gear mechanics** for the movement of bot • Optimized gear ratios and configurations to **maximize bot's power transmission**, resulting in improved overall efficiency • Integrated Arduino-based system for **movement control**, leveraging sensors, and seamless communication interface
Result	• Conducted tests demonstrating bot's impressive climbing capabilities, achieving angles of upto 60° with a **1 N.m** motor

RL Based Optimization Of HVAC Systems | *Association of Mechanical Engineers* — *Jan'23 - Mar'23*

Literature	• Conducted an literature review on RL-based formulations and their application in HVAC, analyzed latest research findings
Model	• Computed optimal state for HVAC systems by leveraging **computational methods** and considering multiple variable factors • Developed a optimization model to **reduce energy consumption and CO2 emissions**, aligning with **sustainability goals** • Evaluated **model performance** on HVAC systems' data, achieving energy savings and substantial CO2 emission reduction

Technical Skills

Languages: Python, C, C++, C#, SQL, R, Bash
Softwares: MATLAB, Solidworks, Ansys, Microsoft Office
Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, OpenCV
Soft Skills: Leadership, Team Work, Communication

Relevant Courses

** Online # Ongoing*

Manufacturing Processes	Nature & Properties of Materials	Thermodynamics	Mechanics of Solids
Engineering Graphics	Theory of Machines	Fluid Mechanics	New Product Development
Fundamentals of Computing	Applied Data Science Lab*	Supervised Machine Learning*	Energy Systems

Positions of Responsibility

Coordinator, Game Development Club | *SnT Council, IIT Kanpur* — *Apr'23 - Present*

Leadership	• Spearheaded a **3-tier team** of **30+** members to organize club events, including workshops, sessions, for campus community • Coordinated with Core Team to organize Semester Projects, for **100+ students**, focusing on topics like Anti-Cheat and AI
Initiatives	• Conducted **interactive workshop** on strategy-game design, attended by diverse audience of **50+** from campus community • Started **"Just Insights"** social media campaign, reaching an audience of **2.5K+** to promote gamedev within the institution
Impact	• Boosted active contributors by **1.6** times, cultivating a more dynamic and collaborative atmosphere within the club

Senior Member, Team Autonomous Underwater Vehicles | *SnT Council, IIT Kanpur* — *Apr'23 - Present*

Team Work	• Collaborated with **multidisciplinary team** on different phases, to develop and optimize a **compact design** for the Anahita • Conducted freshers orientation for **1200+** freshers at SnT Pavilion, delivering a presentation showcasing our work
Initiatives	• Optimized camera casing for **Anahita** (AUV), strategically repositioning it to reduce turbulence and enhance field of view • Proposed new design for torpedo, reducing turbulence and improving accuracy, thereby enhancing overall performance

Extra-Curricular Activities

Design	• **Bronze Medal** at Inter IIT Cult Meet 5.0 in **Design Marathon** under digital art category conducted by IIT Madras • Secured **2nd position** in **Mascot Design** at Galaxy'23 (Inter-hall competition) organized by MnC Council	*2023* *2023*
Case study	• Received **letter of recommendation** as recognition of performance during **Business Analyst Program**, Finlatics • Awarded with **Merit Prize** for completing cases with unique approach at **StrategyCo.Global's Consulting course**	*2023* *2022*